

Faculty Of Engineering and Technology

Electrical and Computer Engineering Department

ENCS3330

Digital Integrated Circuits

Assignment #1

Prepared by : Rahaf Salahat **ID: 1231584**

Instructor: Dr. Khader Mohammad **Section: 2**

Date: 21 – May- 2026

Contents

1-**Inverter Gate** 3

1.1 Schematic & Icon 3

1.2 Layout 3

1.3 Simulation 4

1.4 Discussion: CMOS Inverter Design 4

2-NAND Gate 4

2.1 Schematic & Icon 5

2.2 Layout 6

2.3 Simulation 7

2.4 Discussion: CMOS NAND Design 7

3- NOR Gate 8

3.1 Schematic & Icon 8

3.2 Layout 9

3.3 Simulation 10

3.4 Discussion: CMOS NOR Design 10

Conclusion 12

Table Of Figures

Figure 1: Inverter gate Schematic 3
Figure 2: Layout of Inverter Gate 3
Figure 3:Output Simulation based on Schematic and Layout 4
Figure 4::Schematic of 2 inputs NAND gate 5
Figure 5: NAND gate Icon 5
Figure 6:Layout of NAND Gate 6
Figure 7:Output Simulation based on Schematic and Layout 7
Figure 8 :Schematic of NOR gate 8
Figure 9 : Icon of NOR gate 8
Figure 10: Layout of NOR gate 9
Figure 11:: Output Simulation based on Schematic and Layout 10

1-Inverter Gate

1.1 Schematic & Icon

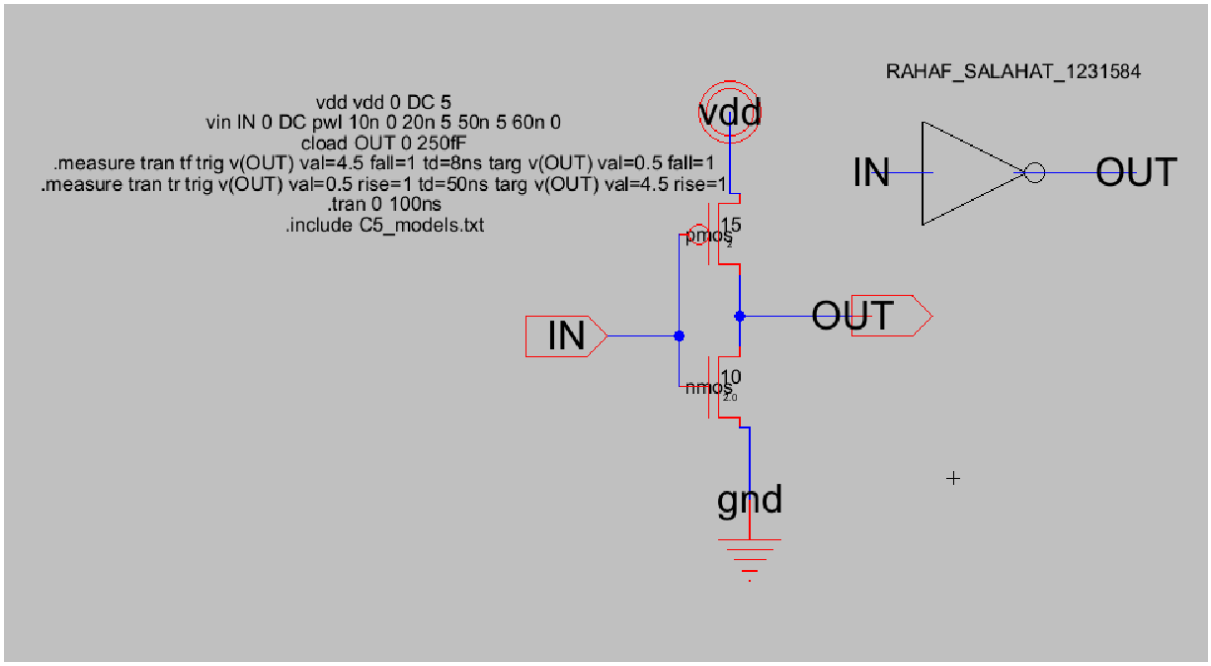


Figure 1: Inverter gate Schematic

1.2 Layout

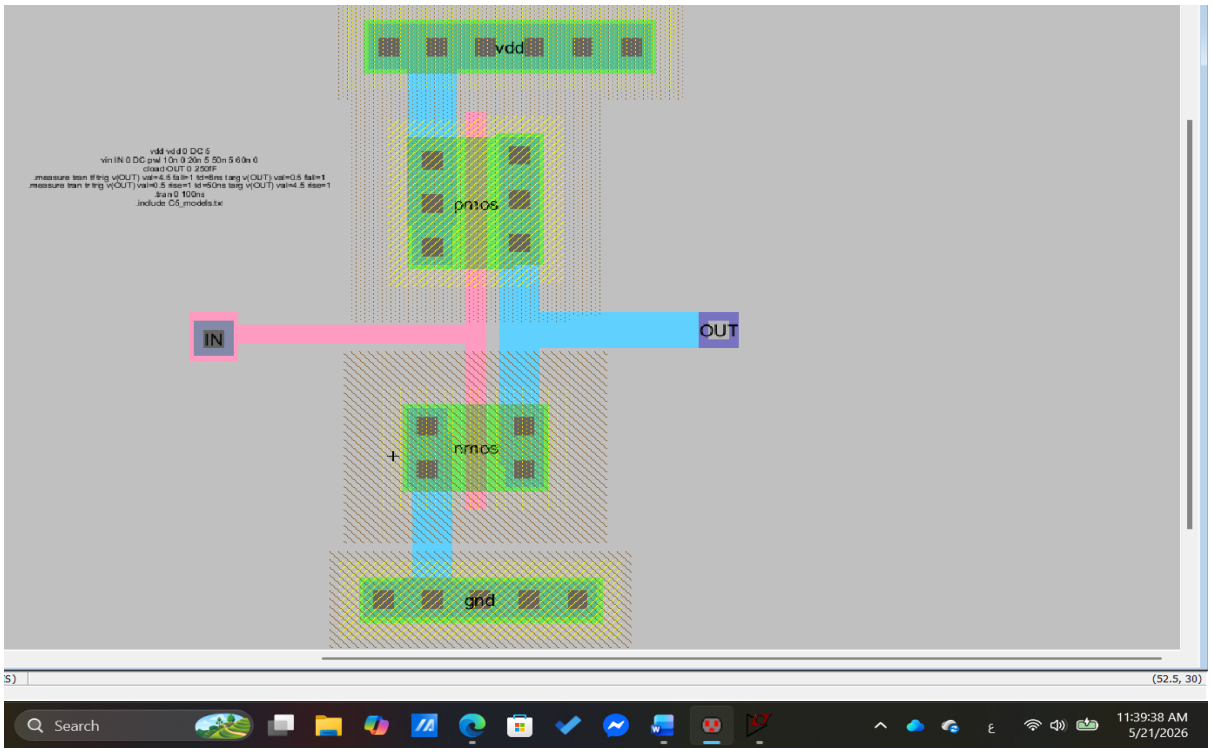


Figure 2: Layout of Inverter Gate

1.3 Simulation

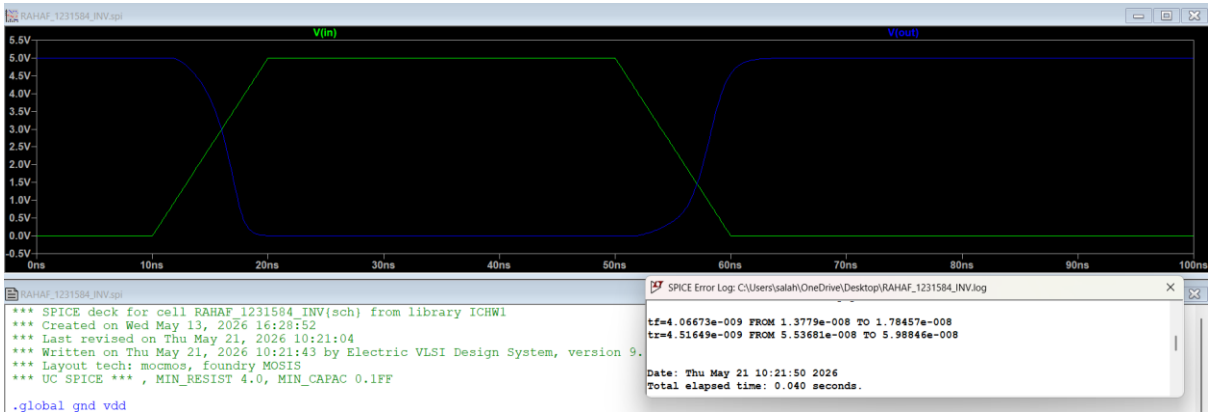


Figure 3:Output Simulation based on Schematic and Layout

It can be noticed that trise and tfall are approximately equal.

1.4 Discussion: CMOS Inverter Design

The CMOS Inverter was successfully verified by matching its logical Schematic with its physical Layout and validating it via LTspice Waveforms. To balance the carrier mobility mismatch between the PMOS (Pull-Up Network) and NMOS (Pull-Down Network), geometric sizing optimization was applied with channel widths of **Wp = 15** and **Wn = 10** over a uniform length of **L = 2**. This intentional ratio (Wp/Wn = 1.5) effectively compensated for the lower intrinsic mobility (mu_p) of holes in the PMOS compared to the higher mobility (mu_n) of electrons in the NMOS. Consequently, this structural scaling equalized the rise and fall times (tr approx tf), establishing a symmetric switching threshold (Vth approx VDD/2) and maximizing both high and low noise margins (N MH and N ML).

The physical layout was fabricated using specialized MOSIS layers under strict DRC and NCC rules. The shared gate was formed using Polysilicon tracks crossing over active diffusion areas, while signal and power routing utilized low-resistivity Metal-1 layers to connect VDD (5V) and gnd (0V). To eliminate substrate body effects and prevent CMOS latch-up, dedicated bulk connections were routed via N-well and P-well contacts, where the N-well contact is adjusted to a minimum dimension (Y-size >= 12) to satisfy width guidelines without encroaching on nearby diffusion boundaries. Interconnections between these overlapping layers were securely linked using optimized Contact plugs.

Finally, transient simulations proved robust rail-to-rail digital inversion. When the input (Vin) transitions to a logical HIGH (5V), the NMOS turns ON and pulls the output (Vout) to a logical LOW (0V). Conversely, when the input drops to LOW (0V), the PMOS switches ON and pulls the output back to HIGH (5V). The post-layout simulation revealed a minor propagation delay caused by intrinsic RC networks from parasitic wire resistances and junction capacitances between overlapping layers. However, the compact layout minimized these parasitics, ensuring sharp switching edges and complete logic functionality.

1-NAND Gate

For the two-input NAND gate, the pull-down network consists of two NMOS transistors in series, which increases the effective channel resistance. To compensate for this effect and maintain equivalent driving capability to the reference inverter, the width of each NMOS transistor was doubled relative to the inverter baseline (10X2). The PMOS transistors are connected in parallel, and therefore their widths were kept equal to the reference inverter PMOS width since only one branch conducts in the worst-case scenario. Accordingly, the NAND gate was precisely sized with **Wp=15** for each parallel PMOS and **Wn=20** for each series NMOS over a uniform channel length of **L=2**.

2.1 Schematic & Icon

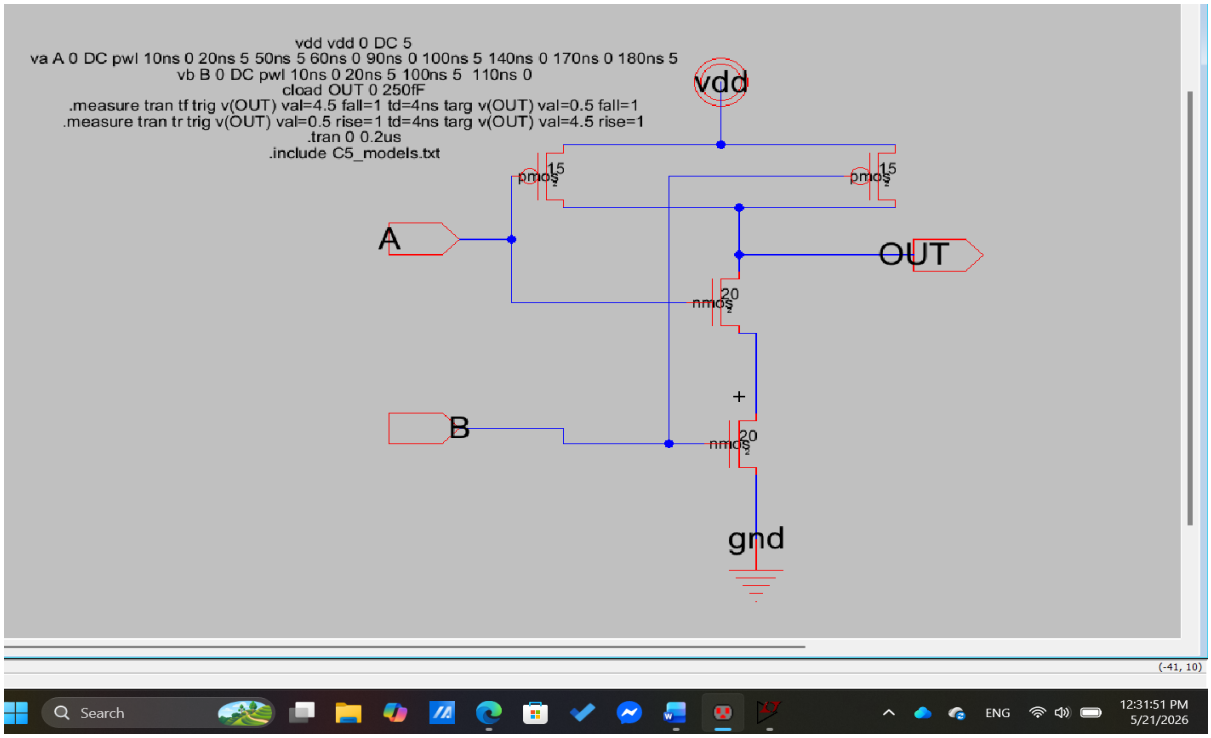


Figure 4:Schematic of 2 inputs NAND gate

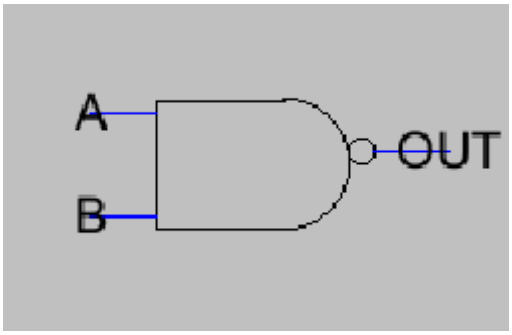


Figure 5: NAND gate Icon

2.2 Layout

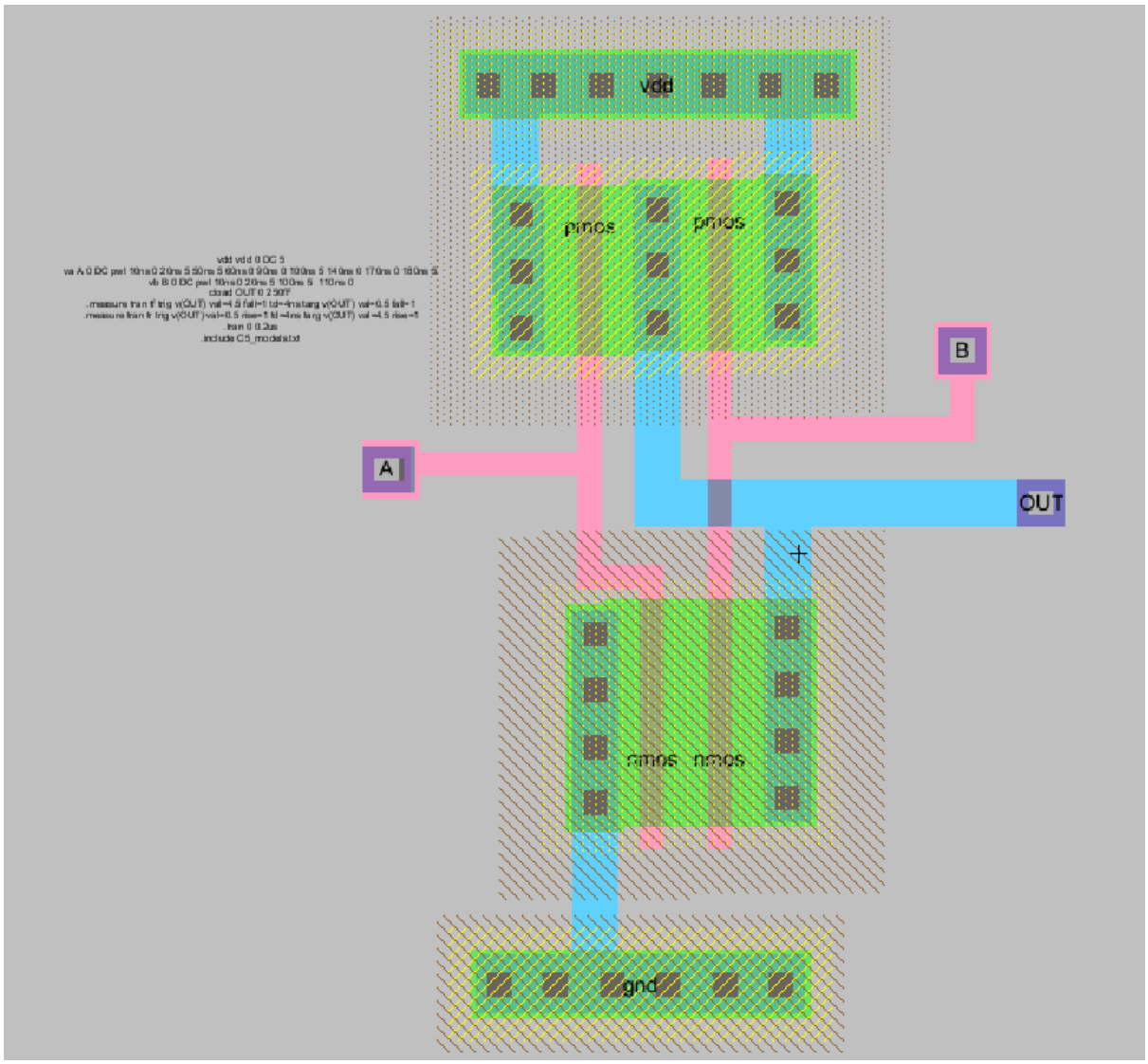


Figure 6:Layout of NAND Gate

2.3 Simulation

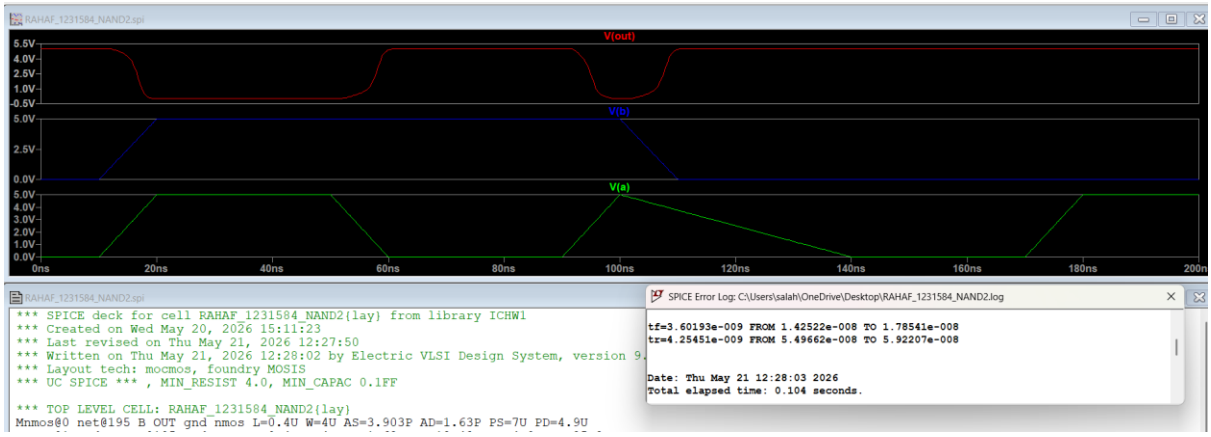


Figure 7:Output Simulation based on Schematic and Layout

It can be noticed that trise and tfall are approximately equal.

2.4 Discussion: CMOS NAND Design

The 2-input CMOS NAND gate was successfully verified by matching its logical Schematic with its physical Layout and validating it via LTspice Waveforms. To balance the carrier mobility mismatch (where NMOS electrons are 2 to 3 times faster than PMOS holes) and compensate for the doubled resistance of the series NMOS network, geometric sizing optimization was applied with channel widths of Wp = 15um and Wn = 20um over a uniform length of L = 2um. This equivalent transistor scaling successfully equalized the pull-up and pull-down driving capabilities, establishing stable noise margins and matching the performance of the reference inverter cell.

The physical layout was fabricated using specialized MOSIS technology layers under strict DRC and NCC rules. The input terminals A and B were routed using vertical Polysilicon tracks crossing over active diffusion regions, while internal connectivity and output routing (OUT) utilized low-resistivity Metal-1 layers to connect VDD (5V) and gnd (0V). To eliminate substrate body effects and prevent CMOS latch-up, dedicated bulk connections were integrated via N-well and P-substrate contacts. Interconnections between overlapping layers were linked securely using optimized Contact plugs.

Finally, transient simulations over an 80ns window proved robust digital NAND functionality. The output voltage drops to a logical LOW (0V) if and only if both inputs A and B are HIGH (5V) simultaneously; for all other states, the output remains HIGH (5V). The post-layout simulation revealed a minor propagation delay caused by intrinsic RC networks from parasitic wire resistances and junction capacitances between overlapping layers. However, the compact layout minimized these parasitics, ensuring sharp switching edges and complete logic functionality without voltage clipping.

2-NOR Gate

3.1 Schematic & Icon

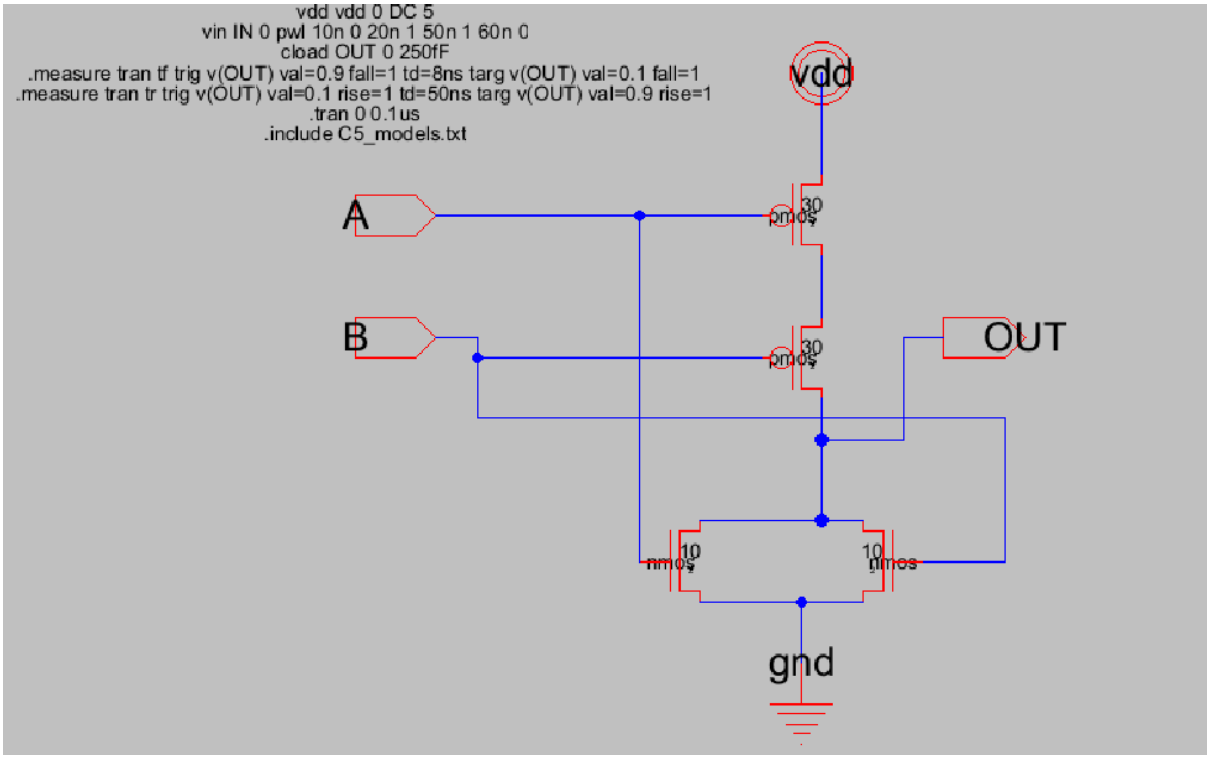


Figure 8 :Schematic of NOR gate

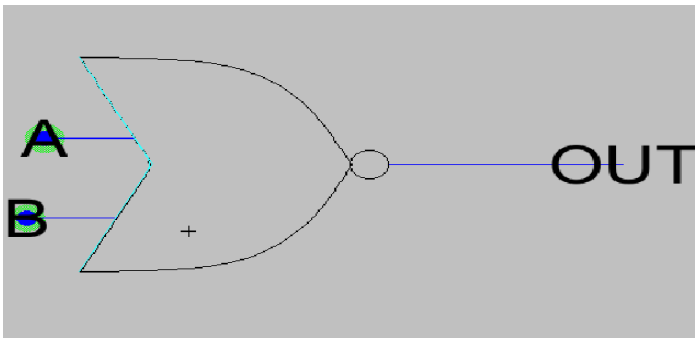


Figure 9 : Icon of NOR gate

3.2 Layout

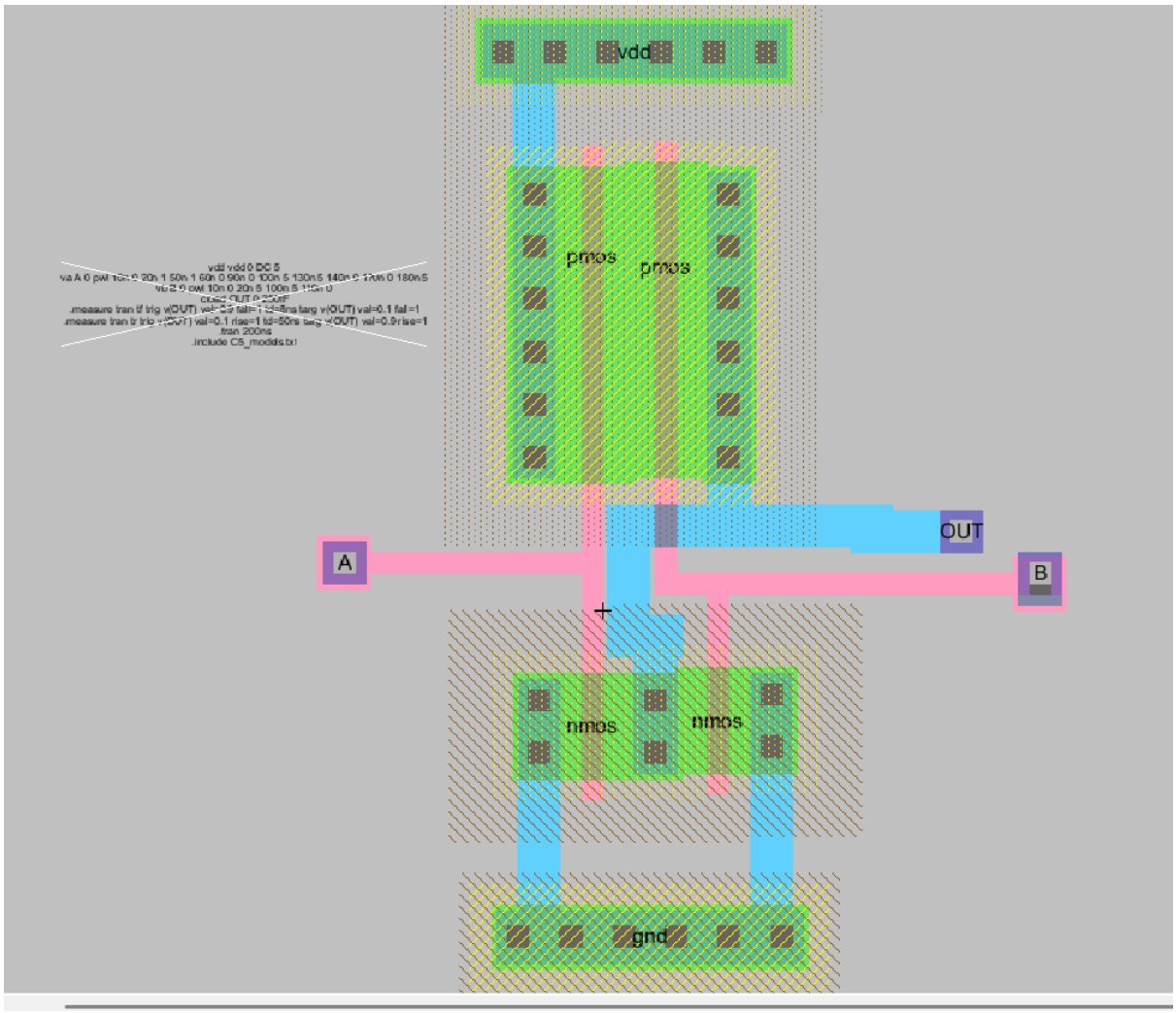


Figure 10: Layout of NOR gate

3.3 Simulation

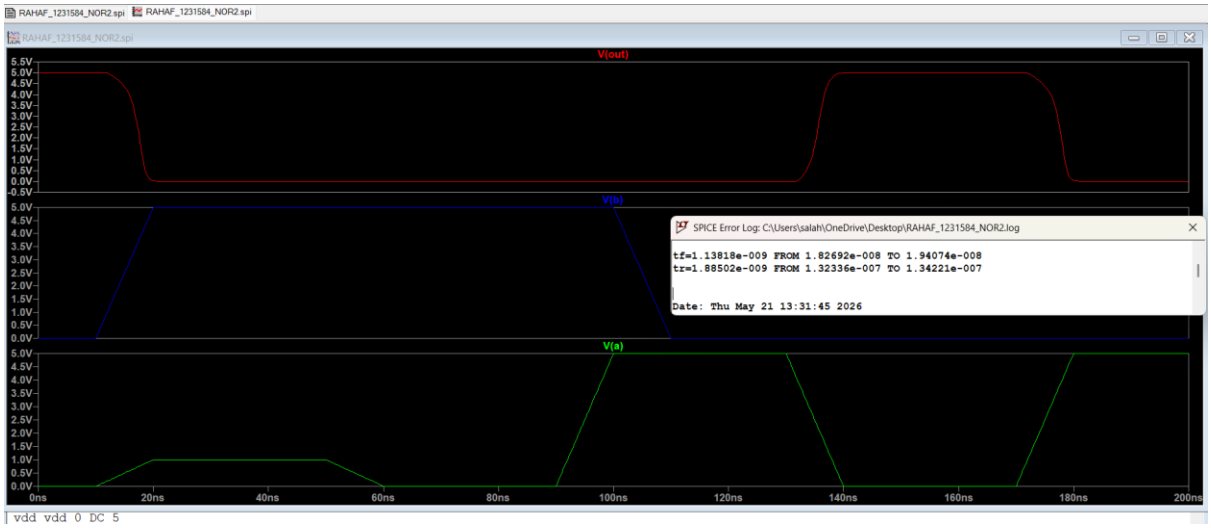


Figure 11:: Output Simulation based on Schematic and Layout

It can be noticed that trise and tfall are approximately equal.

3.4 Discussion: CMOS NOR Design

The 2-input CMOS NOR gate is designed using a complementary structure, where geometric sizing optimization was strictly applied with channel widths of **Wp = 30** for each series PMOS transistor within the Pull-Up Network (PUN), and **Wn = 10** for each parallel NMOS transistor within the Pull-Down Network (PDN) over a uniform length of **L = 2**. This structural arrangement is carefully scaled to match the driving capability of the reference inverter cell; since the series configuration of the PMOS network doubles the charging path resistance, the width of each PMOS was scaled up to **30** (15X2) to maintain symmetric rise and fall transitions. Conversely, because the NMOS transistors are arranged in parallel, each retains the reference width of **10** to provide sufficient discharging speed under worst-case single-branch conduction. This design strictly enforces the NOR truth table, where the output node OUT pulls up to VDDonly when both inputs (A and B) are at Logic 0. If either input shifts to Logic 1, the parallel NMOS paths activate, swiftly discharging the output to gnd.

Looking at the simulation **waveforms**, the dynamic response matches the theoretical expectations perfectly. The output V(OUT) drops directly to 0.0V whenever V(A) or V(B) rises to 5.0V. A slight propagation delay is visible during the low-to-high output transition. This is due to the series configuration of the PMOS transistors, which increases the overall resistance of the charging path. Nevertheless, the signal levels remain sharp, stable, and completely free of unwanted glitches.

From a physical perspective, the layout efficiently translates the schematic into silicon layers. The gates are formed by polysilicon lines crossing the active diffusion regions (N-diffusion for NMOS and P-diffusion for PMOS). Blue Metal 1 routing is utilized to connect the VDD and gnd rails, as well as to combine the drain regions to form the OUT terminal through vertical contact cuts. By minimizing the lengths of these metal wires and grouping the transistors

tightly, parasitic capacitances and resistances are significantly reduced. This optimal layout design directly enhances the switching speed and ensures the stable dynamic performance observed in the transient waveforms.

tightly, parasitic capacitances and resistances are significantly reduced. This optimal layout design directly enhances the switching speed and ensures the stable dynamic performance observed in the transient waveforms.

Conclusion

In this assignment, the design, simulation, and physical layout of three fundamental CMOS logic gates—the Inverter, 2-Input NAND, and 2-Input NOR—were successfully executed and verified. The workflow established a rigorous bridge between theoretical schematic designs and practical silicon implementation. Transient analysis waveforms confirmed the functional integrity of each gate; the outputs accurately matched their respective truth tables under varying pulse inputs (0.0V to 5.0V).

While the Inverter and NAND gates demonstrated sharp transitions and highly symmetric switching behavior, the NOR gate exhibited a more noticeable propagation delay during the low-to-high output transition due to its series PMOS configuration, which inherently increases charging path resistance.

The physical Layout phase highlighted the critical importance of geometric optimization in modern VLSI design. By using Metal 1 (blue) for routing the vdd and gnd power rails and forming low-resistance output nodes, the parasitic resistances and capacitances were kept to a minimum. This careful spatial arrangement is the primary advantage of a well-optimized layout, as it directly enhances switching speed, saves silicon area, and prevents unwanted signal glitches.

However, the design process was not without challenges. Several technical issues were encountered and successfully resolved, including: Design Rule Checking (DRC) violations: Primarily related to minimum spacing and width constraints between active diffusion layers, polysilicon gates, and metal contacts. Layout Versus Schematic (LVS) mismatches: These arose from floating substrate well taps and incorrect node tracking, which required meticulous debugging of the bulk connections to ensure the N-well was tied to vdd and the P-substrate to gnd.

Ultimately, resolving these mismatches achieved perfect LVS clean status. This assignment successfully demonstrated that close integration between tight layout packing and precise schematic sizing is essential to achieving stable, high-performance, and reliable CMOS digital circuits.